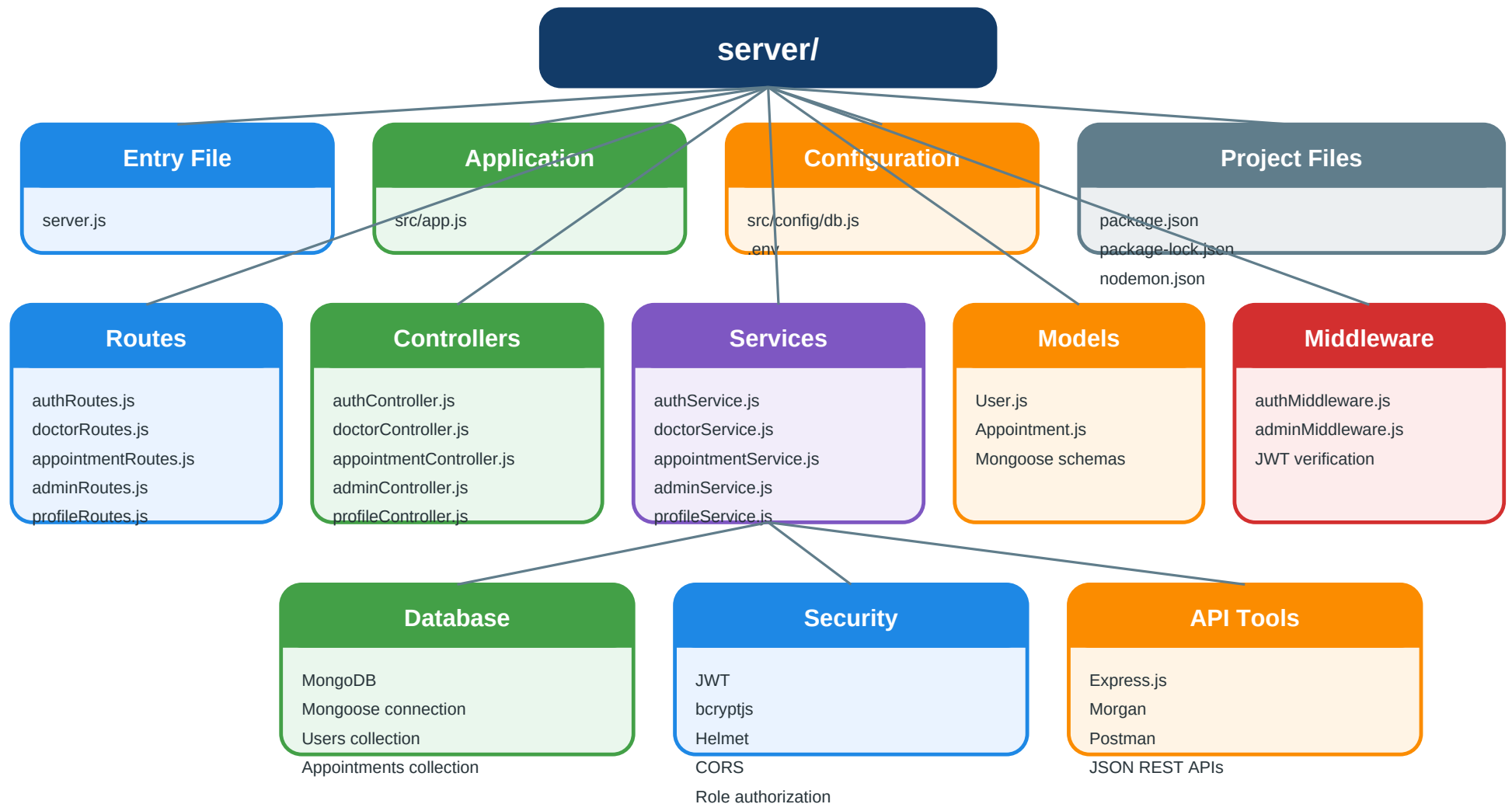


BOOK A DOCTOR SYSTEM - BACKEND STRUCTURE



Backend Folder Responsibilities

Folder / File	Responsibility
server.js	Loads environment variables, connects to MongoDB, imports the Express application, and starts the backend server.
src/app.js	Creates the Express application, registers global middleware, mounts API routes, defines the home endpoint, and handles 404 responses.
src/config	Contains MongoDB connection configuration using Mongoose.
src/routes	Defines REST API endpoints and connects each endpoint to controller functions and middleware.
src/controllers	Receives HTTP requests, calls service functions, handles errors, and returns structured JSON responses.
src/services	Contains business logic, database operations, password hashing, JWT generation, doctor management, and appointment workflow logic.
src/models	Defines Mongoose schemas for User and Appointment documents.
src/middleware	Verifies JWT tokens and restricts administrator-only endpoints.
package.json	Stores project metadata, scripts, and backend dependencies.

Backend Request Flow

Client Request -> Express Route -> Authentication Middleware -> Controller -> Service Layer -> Mongoose Model -> MongoDB -> JSON Response -> React Client

Outcome

The backend is organized using MVC architecture with an additional Service Layer. This structure separates routing, request handling, business logic, database schemas, security, and configuration. As a result, the Book a Doctor backend is modular, reusable, secure, and easier to maintain.